

PHP MySQL PDO Role-Based Authentication System

Tech Stack: PHP + MySQL + PDO + HTML + CSS + JavaScript

This project focuses only on:

- Authentication
 - Role-Based Access Control (RBAC)
 - Route Protection
 - Session Management .
-

1. Project Overview

Create a secure login system where users can log in and access pages based on their role.

Example:

Role	Access
Admin	Full dashboard access
Manager	Manage limited modules
User	View own dashboard

2. Main Features

Authentication

- Register
 - Login
 - Logout
 - Password Hashing
 - Session Management
-

Role-Based Access

- Admin dashboard
- Manager dashboard
- User dashboard
- Restrict unauthorized pages

User Management

(Admin only)

- Create user
 - Edit user
 - Delete user
 - Assign role
 - Activate/Deactivate user
-

3. Roles

Admin

Can:

- Manage users
 - View all dashboards
 - Assign roles
-

Manager

Can:

- View manager dashboard
 - Manage limited data
-

User

Can:

- Login
 - View own dashboard
-

4. Database Design

users table

Column	Type
id	bigint
name	varchar
email	varchar
password	varchar
role	enum/admin-manager-user
status	tinyint
created_at	timestamp

5. Folder Structure

```
project/
├── config/
│   └── database.php
├── controllers/
│   ├── AuthController.php
│   ├── UserController.php
│   └── DashboardController.php
├── middleware/
│   ├── auth.php
│   └── role.php
├── views/
│   ├── auth/
│   ├── admin/
│   ├── manager/
│   └── user/
├── public/
│   ├── css/
│   └── js/
├── includes/
│   ├── header.php
│   └── footer.php
├── routes/
│   └── web.php
└── index.php
```

6. Authentication Flow

Register
↓

Password Hash
↓
Store User
↓
Login
↓
Verify Password
↓
Create Session
↓
Check Role
↓
Redirect Dashboard

7. Registration Module

Fields

- Name
 - Email
 - Password
 - Confirm Password
 - Role
-

Validation

- Email unique
 - Password minimum 8 characters
 - Required fields
-

8. Login Module

Features

- Email login
 - Password verify
 - Session create
 - Redirect by role
-

Example Redirect

```
if($user['role'] == 'admin'){
    header("Location: admin/dashboard.php");
}
elseif($user['role'] == 'manager'){
    header("Location: manager/dashboard.php");
}
else{
    header("Location: user/dashboard.php");
}
```

9. Role Middleware

Purpose

Prevent unauthorized access.

Example

```
session_start();

if($_SESSION['role'] != 'admin'){
    die("Access Denied");
}
```

10. Auth Middleware

Purpose

Check user logged in or not.

Example

```
if(!isset($_SESSION['user_id'])){
    header("Location: login.php");
}
```

11. PDO Database Connection

database.php

```
<?php

$host = "localhost";
$dbname = "rbac_system";
$username = "root";
$password = "";

try{
    $pdo = new PDO(
        "mysql:host=$host;dbname=$dbname",
        $username,
        $password
    );

    $pdo->setAttribute(
        PDO::ATTR_ERRMODE,
        PDO::ERRMODE_EXCEPTION
    );

}catch(PDOException $e){
    die("Connection Failed");
}
```

12. Password Hashing

Store Password

```
$hashPassword = password_hash($password, PASSWORD_BCRYPT);
```

Verify Password

```
password_verify($password, $user['password']);
```

13. Session Management

Store Login Session

```
$_SESSION['user_id'] = $user['id'];
$_SESSION['name'] = $user['name'];
$_SESSION['role'] = $user['role'];
```

14. Admin Module

Admin Can

- Create users

- Edit users
 - Delete users
 - Change roles
 - View all users
-

15. Dashboard Pages

Admin Dashboard

- Total users
 - Role statistics
 - User management
-

Manager Dashboard

- Team overview
 - Reports
-

User Dashboard

- Profile
 - Account details
-

16. Security Features

PDO Prepared Statements

```
$stmt = $pdo->prepare(  
    "SELECT * FROM users WHERE email = ?"  
);
```

```
$stmt->execute([$email]);
```

Password Encryption

- `password_hash()`

- `password_verify()`
-

Session Security

- Session destroy on logout
 - Session regenerate ID
-

XSS Protection

```
echo htmlspecialchars($name);
```

17. Frontend Pages

Public

- Login
 - Register
-

Protected

- Dashboard
 - User Management
-

18. Final Project Flow

```
User Register
↓
Login
↓
Session Create
↓
Check Role
↓
Redirect Dashboard
↓
Access Allowed Pages
```